Theory of Manifold Interfaces

Regularized Effective Physical Layer and Interface-Field Program

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Abstract

This paper presents the physical layer of the Theory of Manifold Interfaces as a regularized effective interface-field research program. The central formula states that geometry determines the domain of admissible physical transitions, quantum theory determines a probability distribution over this domain, and an interface field mediates the transition from distributed possibilities to stable physical structures. Formally, the construction is expressed as

$$g \Rightarrow \mathrm{Adm}_c(g) \Rightarrow \mathcal{H}_I(g) \Rightarrow \Psi_I \Rightarrow \mathrm{Dist}_{\mathrm{poss}}(\Psi_I; g) \Rightarrow \chi_I \Rightarrow \mathrm{Struct}_I\,.$$

The paper introduces a minimal interface field χ_I, an effective action, a regularized space of geometries, an interface distance between geometry classes, a physical normalization of the interface decoherence coefficient, consistency limits recovering general relativity and the Standard Model, and an observable matrix for quantum-geometric decoherence, the dark sector, black holes, and an interface-field architecture for unification. The paper does not claim to provide a completed fundamental theory or a final Theory of Everything. Its result is a rigorous effective model with explicit definitions, consistency limits, falsifiable hypotheses, and open problems.

Keywords: Theory of Manifold Interfaces; interface field; admissible transitions; quantum geometry; decoherence; dark sector; black holes; Standard Model; unification; effective field theory.

Contents

1 Introduction

Fundamental physics contains several powerful but conceptually distinct descriptions. General relativity describes spacetime through the metric $g_{\mu\nu}$ and relates it to energy and momentum through Einstein's equation. Quantum theory describes physical states as vectors, functionals, or density matrices that encode distributions of possible outcomes. The Standard Model describes gauge interactions and fermionic fields. Cosmology and black-hole physics add further unresolved issues: the dark sector, horizons, entropy, the information problem, and the quantum nature of geometry.

The Theory of Manifold Interfaces proposes a common language connecting three operations: admissibility, distribution, and structuring. In the physical layer of TMI, geometry answers which transitions are physically admissible; quantum theory answers how possibilities are distributed; the interface field answers how a distribution of possibilities becomes a stable structure.

The central chain is

$$\boxed{g \Rightarrow \mathrm{Adm}_c(g) \Rightarrow \mathcal{H}_I(g) \Rightarrow \Psi_I \Rightarrow \mathrm{Dist}_{\mathrm{poss}}(\Psi_I; g) \Rightarrow \chi_I \Rightarrow \mathrm{Struct}_I\,.}$$

In words,

$$\boxed{\text{geometry determines admissibility; \quad quantum theory determines distribution; \quad th}}$$

The aim of this paper is to assemble this chain into a coherent physical layer: to define the basic objects, write a minimal action, derive the equations of motion, identify the main regimes, define observable quantities, state limitations, and formulate the status of the model. The central methodological point is that TMI-Physics is treated as an effective interface-field program, not as a completed fundamental theory. This prevents overstatement and makes the proposal suitable for further refinement.

2 Relation to Previous TMI Materials

This publication is a synthetic step in the physical branch of the Theory of Manifold Interfaces. Previous modules treated the mathematical abstraction of TMI, a cautious interface approach to quantum gravity, quantum-geometric decoherence, the dark sector as an interface field, black holes as causal-informational interfaces, and the interface-field architecture of unification. In this paper these topics are not treated as isolated branches. They are unified as regimes of one minimal system:

$$\boxed{(g_{\mu\nu}, \Psi, \chi_I, \mathbb{A}_I, \mathcal{G}_I) \Rightarrow (QG, Dark, BH, SM, Gen, ToE).}$$

Here QG denotes the quantum-geometric regime, $Dark$ the dark sector, BH black holes, SM the Standard Model as a limiting sector, Gen the fermion-generation structure, and ToE the unification program through a universal interface connection.

3 Geometric Admissibility of Physical Transitions

Let $(\mathcal{M}_U, g)$ be a spacetime with Lorentzian metric. At each point $x \in \mathcal{M}_U$ there is a tangent space $T_x\mathcal{M}_U$. Define the future interface causal cone by

$$\boxed{\mathcal{C}^+_{I,g}(x) := \{v \in T_x\mathcal{M}_U \setminus \{0\} \mid g_x(v,v) \leq 0,\ v \text{ is future-directed}\}.}$$

This is not a new object beyond general relativity. It is the standard future causal cone interpreted as the set of locally admissible directions of physical interface transition:

$$\boxed{\mathcal{C}^+_{I,g}(x) \equiv \mathcal{C}^+_g(x).}$$

Its boundary is given by

$$g_x(v,v) = 0,$$

and therefore

$$\boxed{\partial\mathcal{C}^+_{I,g}(x) \equiv \mathcal{C}^+_{\text{light},g}(x).}$$

Thus the light cone is interpreted in TMI as the boundary of admissible causal interface transition.

Let $J^+_g(x)$ denote the causal future of x. The set of causally admissible transitions is defined by

$$\boxed{\text{Adm}_c(g) := \{(x,y) \in \mathcal{M}_U \times \mathcal{M}_U \mid y \in J^+_g(x)\}.}$$

This fixes the first layer of the theory:

$$\boxed{g \Rightarrow \mathcal{C}^+_{I,g}(x) \Rightarrow J^+_g(x) \Rightarrow \text{Adm}_c(g).}$$

The physical meaning is that the metric determines not only distances and curvature but also the structure of physically admissible transitions.

4 Quantum Distribution over the Admissible Domain

Geometry determines the admissible domain, but it does not determine which admissible transition is realized. A quantum layer is therefore introduced. Consider the measurable space

$$(\text{Adm}_c(g), \mathcal{B}_g, \mu_g),$$

where $\mathcal{B}_g$ is a σ-algebra of measurable subsets and μ_g is an effective or regularized measure. A complete fundamental measure on the general space of transitions or geometries is not assumed to be constructed in this paper. Four levels are allowed:

$$\boxed{\mu_g \in \{\mu_g^{disc}, \mu_g^{mini}, \mu_g^{cyl}, \mu_g^{PI}\}.}$$

Here μ_g^{disc} is a discrete measure, μ_g^{mini} a minisuperspace measure, μ_g^{cyl} a cylindrical measure on a regularized projective system, and μ_g^{PI} an effective path-integral measure.

The Hilbert space of interface possibilities is

$$\boxed{\mathcal{H}_I(g) := L^2(\mathrm{Adm}_c(g), \mathcal{B}_g, \mu_g).}$$

If $\psi_I \in \mathcal{H}_I(g)$, then the probability distribution on admissible transitions is

$$\boxed{d\mathbb{P}_{\Psi,g}(\Phi) = |\psi_I(\Phi)|^2 \, d\mu_g(\Phi), \qquad \Phi \in \mathrm{Adm}_c(g).}$$

The second layer is therefore

$$\boxed{\mathrm{Adm}_c(g) \Rightarrow \mathcal{H}_I(g) \Rightarrow \Psi_I \Rightarrow \mathrm{Dist}_{\mathrm{poss}}(\Psi_I; g).}$$

5 Interface Structuring of Possibilities

A distribution of possibilities is not identical to a realized structure. The interface field is introduced to describe the transition from distribution to stable physical regime. In the minimal model,

$$\boxed{\chi_I : \mathcal{M}_U \to \mathbb{R},}$$

and in a more general geometric form,

$$\boxed{\chi_I \in \Gamma(\mathcal{E}_\chi), \qquad \mathcal{E}_\chi \to \mathcal{M}_U.}$$

Interface structuring is defined as a map

$$\boxed{\mathcal{S}_I : \mathrm{Dist}_{\mathrm{poss}}(\Psi_I; g) \longrightarrow \mathrm{Struct}_I \, .}$$

More fully,

$$\boxed{\mathrm{Struct}_I = \mathcal{S}_I(g, \Psi_I, \chi_I, \mathrm{Adm}_c(g)).}$$

In the simplest two-geometry model,

$$|\Psi_g\rangle = c_1|g_1\rangle + c_2|g_2\rangle, \qquad |c_1|^2 + |c_2|^2 = 1,$$

coherence between the alternatives is represented by an off-diagonal density-matrix element. Interface structuring is modeled by

$$\boxed{|\rho_{12}(t)| = e^{-\Gamma_I t}|\rho_{12}(0)|.}$$

A natural degree of structuring is

$$\boxed{\mathrm{Struct}_I(t) := 1 - \frac{|\rho_{12}(t)|}{|\rho_{12}(0)|} = 1 - e^{-\Gamma_I t}.}$$

6 Minimal Effective Action

The physical layer of TMI is encoded by the minimal effective action

$$\boxed{S_{TMI}^{min} = S_{GR} + S_{SM} + S_\chi + S_{mix} + S_{dec}^{eff}.}$$

In local form,

$$\boxed{S_{TMI}^{min} = \int_{\mathcal{M}_U} \sqrt{-g}\left[\frac{c^4}{16\pi G}(R - 2\Lambda) + \mathcal{L}_{SM} + \mathcal{L}_\chi + \mathcal{L}_{mix}\right] d^4x + S_{dec}^{eff}.}$$

Here S_{dec}^{eff} is not a fundamental local action in the usual sense. It is an effective term for open quantum dynamics and interface-induced decoherence.

The minimal Lagrangian of the interface field is

$$\boxed{\mathcal{L}_\chi = -\frac{1}{2}g^{\mu\nu}\nabla_\mu\chi_I\nabla_\nu\chi_I - V(\chi_I) - \frac{1}{2}\xi_I R\chi_I^2.}$$

The potential is

$$\boxed{V(\chi_I) = V_0 + \frac{1}{2}m_I^2\chi_I^2 + \frac{\beta_I}{4}\chi_I^4.}$$

Stability of the minimal model requires the potential to be bounded from below. A sufficient simple condition is

$$\boxed{\beta_I > 0, \qquad V(\chi_I) \geq 0 \text{ in the physically admissible domain.}}$$

Allowed mixings must preserve the gauge structure of the Standard Model. Thus, in the minimal version χ_I is chosen as a gauge singlet:

$$\boxed{\chi_I \sim (\mathbf{1}, \mathbf{1}, 0) \quad \text{with respect to } SU(3) \times SU(2) \times U(1)_Y.}$$

An example of a gauge-safe mixing term is

$$\boxed{\mathcal{L}_{mix}^{gauge} = -\frac{1}{4}\sum_a \zeta_a \frac{\chi_I^2}{M_I^2} F_{\mu\nu}^a F_a^{\mu\nu}.}$$

7 Equations of Motion of the Minimal Model

Variation with respect to the metric gives the generalized geometric equation

$$\boxed{G_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4}\left(T_{\mu\nu}^{SM} + T_{\mu\nu}^{\chi} + T_{\mu\nu}^{mix}\right) + \mathcal{D}_{\mu\nu}^{eff}.}$$

The term $\mathcal{D}_{\mu\nu}^{eff}$ denotes a possible effective correction from open quantum dynamics. If the backreaction of decoherence on geometry is not included, one may set $\mathcal{D}_{\mu\nu}^{eff} = 0$.

Variation with respect to χ_I gives the interface-field equation

$$\boxed{\Box_g\chi_I - (m_I^2 + \xi_I R)\chi_I - \beta_I\chi_I^3 = -J_I.}$$

The source J_I encodes the excitation of the interface field by quantum-geometric or matter configurations. In the QG regime the minimal form is

$$\boxed{J_I = \alpha_I |c_1|^2 |c_2|^2 d_I^2(g_1, g_2).}$$

Open quantum dynamics is written as

$$\boxed{\frac{d\rho}{dt} = -\frac{i}{\hbar}[\widehat{H}_{eff}, \rho] + \mathcal{L}_{dec}^I[\rho].}$$

The minimal decoherence superoperator is

$$\boxed{\mathcal{L}_I[\rho] = -\frac{1}{2}\kappa_I[\widehat{D}_I, [\widehat{D}_I, \rho]].}$$

Gauge fields obey modified equations of the form

$$\boxed{D_\mu \left[\left(1 + \zeta_a \frac{\chi_I^2}{M_I^2} \right) F_a^{\mu\nu} \right] = J_a^\nu.}$$

8 Regularized Space of Geometries

The quantum-geometric regime requires moving from a finite superposition of geometries to a space of geometric possibilities. Let $\mathcal{G}(\mathcal{M}_U)$ be the space of admissible Lorentzian metrics. A physical geometry is not a coordinate representation of a metric but a diffeomorphism class:

$$\boxed{[g] = \{\varphi^* g \mid \varphi \in \mathrm{Diff}(\mathcal{M}_U)\}.}$$

The space of physical geometries is

$$\boxed{\mathfrak{G} := \mathcal{G}(\mathcal{M}_U)/\,\mathrm{Diff}(\mathcal{M}_U).}$$

A complete measure on $\mathfrak{G}$ is not claimed to be constructed. Instead, a finite-dimensional regularization is introduced:

$$\boxed{\mathcal{G}_N \subset \mathcal{G}(\mathcal{M}_U), \qquad \mathfrak{G}_N = \mathcal{G}_N/\,\mathrm{Diff}_N\,.}$$

The working Hilbert space of geometries is

$$\boxed{\mathcal{H}_G^{eff} = L^2(\mathfrak{G}_N, \mathcal{B}_N, \mu_G^{cyl}).}$$

Here μ_G^{cyl} is a regularized cylindrical measure. It is not declared to be a final fundamental measure, but it provides a computable hierarchy of approximations.

The interface distance between physical geometries must compare equivalence classes rather than coordinate representatives:

$$\boxed{d_I([g_1], [g_2]).}$$

A general effective form is

$$\boxed{d_I([g_1],[g_2]) = \frac{1}{\mathcal{N}_I}\inf_{\varphi\in\mathrm{Diff}(\mathcal{M}_U)}\left[\int_\Omega w_I(x)\|g_1-\varphi^* g_2\|^2_{g_0}dV_{g_0}\right]^{1/2}.}$$

In the minisuperspace model with scale factor a,

$$\boxed{d_I(g(a_1),g(a_2)) = \left|\ln\frac{a_1}{a_2}\right|.}$$

9 Physical Normalization of the Decoherence Coefficient

The QG module uses the observable excess

$$\boxed{\Delta\Gamma_I = \kappa_I x_I,}$$

where

$$\boxed{x_I = |c_1|^2|c_2|^2 d^2_{I,N}([g_1],[g_2]).}$$

If x_I is dimensionless, then κ_I has dimensions of inverse time. A physically natural normalization is therefore given by the characteristic frequency of the interface field:

$$\boxed{\omega_I = \frac{m_I c^2}{\hbar}.}$$

Then

$$\boxed{\kappa_I = \frac{m_I c^2}{\hbar}\mathcal{K}_I.}$$

In the minimal effective model,

$$\boxed{\mathcal{K}_I = \lambda_I\alpha_I^2,}$$

and a more general expression may include curvature, self-interaction, and energy-scale corrections:

$$\boxed{\kappa_I = \frac{m_I c^2}{\hbar}\lambda_I\alpha_I^2\,\mathcal{F}_I\left(\frac{\xi_I R}{m_I^2},\beta_I,\frac{E}{M_I},\ldots\right).}$$

Thus κ_I is no longer a completely arbitrary parameter: it is tied to the scale of the interface field and to dimensionless couplings.

10 Quantum Dynamics of the Metric in the Regularized Model

The quantum state of geometry is now considered as a function on the space of physical geometries:

$$\boxed{\Psi_G([g],t) \in \mathcal{H}_G^{eff}.}$$

Normalization is

$$\boxed{\int_{\mathfrak{G}_N} |\Psi_G([g],t)|^2 d\mu_G^{cyl}([g]) = 1.}$$

An effective dynamical equation may be written as

$$\boxed{i\hbar\frac{\partial}{\partial t}\Psi_G([g],t) = \widehat{H}_G^{eff}\Psi_G([g],t).}$$

The parameter t must be understood cautiously: it is an effective or internal time associated with a chosen regularization, not a complete solution of the problem of time in quantum gravity. A more covariant form is a constraint

$$\boxed{\widehat{\mathcal{H}}_{TMI}\Psi = 0.}$$

For open dynamics it is more convenient to use the density matrix:

$$\boxed{\frac{d\rho_G}{dt} = -\frac{i}{\hbar}[\widehat{H}_G^{eff},\rho_G] - \frac{1}{2}\kappa_I[\widehat{D}_I,[\widehat{D}_I,\rho_G]].}$$

If $\widehat{D}_I$ distinguishes geometries by $d_{I,N}$, then

$$\boxed{\Gamma_I([g_1],[g_2]) = \kappa_I d_{I,N}^2([g_1],[g_2]).}$$

Therefore,

$$\boxed{\rho_G([g_1],[g_2];t) = e^{-\Gamma_I([g_1],[g_2])t}\rho_G([g_1],[g_2];0).}$$

A classical metric appears as a decohered limit of a quantum distribution of geometries:

$$\boxed{\rho_G([g_1],[g_2]) \to 0 \quad ([g_1]\neq[g_2]) \Rightarrow g_{\mu\nu}^{cl}.}$$

11 Recovery of the Standard Model and Projections of Interactions

A safe structure of the universal interface group is

$$\boxed{G_I = G_{\text{grav}} \ltimes G_{SM} \ltimes G_\chi,}$$

where

$$\boxed{G_{SM} = SU(3)\times SU(2)\times U(1)_Y, \qquad G_{\text{grav}} \supset \text{Spin}(1,3).}$$

The algebra of the minimal model is

$$\boxed{\mathfrak{g}_I = \mathfrak{g}_{\text{grav}} \oplus \mathfrak{su}(3)\oplus\mathfrak{su}(2)\oplus\mathfrak{u}(1)\oplus\mathfrak{g}_\chi.}$$

The universal interface connection is

$$\boxed{\mathbb{A}_I = \omega^{ab}J_{ab} + G^A T_A + W^i T_i + BY + A_\chi^r \Xi_r.}$$

The universal curvature is

$$\boxed{\mathbb{F}_I = d\mathbb{A}_I + \mathbb{A}_I \wedge \mathbb{A}_I.}$$

Introduce algebra projections

$$\boxed{\pi_a : \mathfrak{g}_I \to \mathfrak{g}_a, \qquad a \in \{G, 3, 2, 1, \chi\}.}$$

They satisfy

$$\boxed{\pi_a^2 = \pi_a, \quad \pi_a \pi_b = 0 \ (a \neq b), \qquad \sum_a \pi_a = \mathrm{id}_{\mathfrak{g}_I}.}$$

Then

$$\boxed{F_a = \pi_a(\mathbb{F}_I).}$$

The statement that interactions are projections of a universal interface curvature is therefore given a precise meaning.

The electromagnetic sector appears after electroweak symmetry breaking:

$$\boxed{SU(2) \times U(1)_Y \to U(1)_{EM}.}$$

Thus

$$\boxed{A_\mu^{EM} = \sin\theta_W W_\mu^3 + \cos\theta_W B_\mu.}$$

The safe Standard Model limit is

$$\boxed{g_{\mu\nu} = g_{\mu\nu}^{fixed}, \quad \chi_I \to 0, \quad S_{mix} \to 0 \Rightarrow S_{TMI}^{min} \to S_{SM}.}$$

With gravity,

$$\boxed{\chi_I \to 0, \quad S_{mix} \to 0 \Rightarrow S_{TMI}^{min} \to S_{GR} + S_{SM}.}$$

12 Fermion Generations as Stable Interface Modes

The Standard Model contains three generations of fermions but does not derive their number from first principles. TMI introduces an internal generation space $\mathcal{H}_{gen}$ and a generation-structuring interface operator

$$\boxed{\mathcal{G}_I : \mathcal{H}_{gen} \to \mathcal{H}_{gen}.}$$

The spectral problem is

$$\boxed{\mathcal{G}_I \psi_f = \lambda_f \psi_f.}$$

A fermionic state has the form

$$\boxed{\Psi_f \in \mathcal{H}_{SM} \otimes \mathcal{H}_{gen}, \qquad \Psi_f = \psi_{SM} \otimes \psi_{gen}^{(f)}.}$$

The generation operator must not destroy the gauge structure:

$$\boxed{[\mathcal{G}_I, G_{SM}] = 0.}$$

Physical generations are identified with stable modes:

$$\boxed{f \leftrightarrow \psi_f \in \mathrm{Spec}_{stable}(\mathcal{G}_I).}$$

The weak result is that generations are formalized as stable interface modes. The strong result remains open:

$$\boxed{|\,\mathrm{Spec}_{stable}(\mathcal{G}_I)| = 3.}$$

Fermion masses may depend on the interface mode:

$$\boxed{m_f = \frac{v_H}{\sqrt{2}}\left[y_f^{SM} + \eta_f \frac{\lambda_f \langle \chi_I \rangle}{M_I}\right].}$$

Generation mixing may arise from non-diagonality of the generation operator:

$$\boxed{\Delta Y_{ij}^I = \eta_{ij} \frac{\langle \psi_i | \mathcal{G}_I | \psi_j \rangle}{M_I} \langle \chi_I \rangle.}$$

13 Physical Regimes of the Minimal System

The minimal system contains four key objects:

$$\boxed{g_{\mu\nu}, \qquad \Psi, \qquad \chi_I, \qquad \mathbb{A}_I.}$$

They generate the main physical regimes.

13.1 Quantum-Geometric Regime

The QG regime describes the suppression of coherence between geometric alternatives:

$$\boxed{\Delta\Gamma_I = \kappa_I x_I, \qquad x_I = |c_1|^2 |c_2|^2 d_I^2([g_1],[g_2]).}$$

The observable effect is excess geometric decoherence.

13.2 Dark Sector

The dark sector is described as an effective contribution of the interface field:

$$\boxed{T_{\mu\nu}^{dark} = T_{\mu\nu}^I[\chi_I, g].}$$

The decomposition

$$\boxed{\chi_I(x,t) = \chi_0(t) + \delta\chi(x,t)}$$

gives two regimes:

$$\boxed{\chi_0(t) \Rightarrow DE, \qquad \delta\chi(x,t) \Rightarrow DM.}$$

For the background mode,

$$\boxed{w_I = \frac{\frac{1}{2}\dot{\chi}_0^2 - V(\chi_0)}{\frac{1}{2}\dot{\chi}_0^2 + V(\chi_0)}.}$$

If $V(\chi_0) \gg \frac{1}{2}\dot{\chi}_0^2$, then $w_I \approx -1$.

13.3 Black Holes

In TMI a black hole is a one-way causal-informational interface. For a Schwarzschild black hole,

$$\boxed{r_s = \frac{2GM}{c^2}.}$$

The horizon $r = r_s$ is the boundary of admissible outward causal transitions:

$$\boxed{\mathcal{H}_g = \partial\,\mathrm{Adm}_{out}\,.}$$

Inside the horizon the classical outward channel is forbidden:

$$\boxed{r < r_s \Rightarrow \mathrm{Adm}_{out}^{class} = \varnothing.}$$

The quantum outward channel is associated with Hawking radiation. The temperature

$$\boxed{T_H = \frac{\hbar c^3}{8\pi G M k_B}}$$

is interpreted as the intensity of the quantum outward interface channel.

13.4 Unification Architecture

The ToE regime is understood cautiously: it is not a completed Theory of Everything but an interface-field program toward unification. The four interactions are described as projections of a universal interface curvature:

$$\boxed{\mathbb{F}_I \Rightarrow (F_G, F_3, F_2, F_1, F_\chi), \qquad F_a = \pi_a(\mathbb{F}_I).}$$

Interface corrections to gauge couplings have the form

$$\boxed{\frac{\Delta g_a}{g_a} \approx -\frac{1}{2}\zeta_a \frac{\chi_I^2}{M_I^2}.}$$

14 Observable Matrix and Falsification Criteria

The physical layer becomes a testable program only if observable outputs are specified. Define the matrix

$$\boxed{\mathcal{T}_{TMI}^{phys} = \left(\Delta\Gamma_I, w_I(z), G_{eff}(k,z), \rho_{corr}, \Delta_a(E)\right).}$$

Here:

- $\Delta\Gamma_I = \Gamma_{obs} - \Gamma_{std}$ is excess geometric decoherence;
- $w_I(z)$ is a possible deviation of dark energy from ideal $w = -1$;

- $G_{eff}(k,z)$ is a possible effective gravitational coupling in the dark sector;
- ρ_{corr} is a correlation contribution to black-hole radiation;
- $\Delta_a(E)$ is an interface correction to gauge couplings.

For the QG regime,

$$\boxed{H_0^{QG} : \Delta\Gamma_I = 0,} \qquad \boxed{H_{TMI}^{QG} : \Delta\Gamma_I = \kappa_I x_I.}$$

For the dark sector,

$$\boxed{H_0^{Dark} : w = -1,\ DM \text{ and } DE \text{ are independent},}$$

$$\boxed{H_{TMI}^{Dark} : DM \text{ and } DE \text{ are regimes of one field } \chi_I.}$$

For black holes,

$$\boxed{H_0^{BH} : \rho_{rad} = \rho_{thermal},} \qquad \boxed{H_{TMI}^{BH} : \rho_{rad} = \rho_{thermal} + \rho_{corr}.}$$

For unification,

$$\boxed{H_0^{ToE} : \Delta_a(E) = 0,} \qquad \boxed{H_{TMI}^{ToE} : \Delta_a(E) = \zeta_a \frac{\chi_I^2(E)}{M_I^2}.}$$

15 Consistency Conditions

The physical layer is admissible only if known theories are recovered in the appropriate limits.

15.1 General Relativity Limit

If

$$\boxed{\chi_I \to 0, \quad S_{mix} \to 0, \quad S_{dec}^{eff} \to 0,}$$

then

$$\boxed{S_{TMI}^{min} \to S_{GR} + S_{matter}.}$$

With no matter,

$$\boxed{S_{TMI}^{min} \to S_{GR}.}$$

15.2 Quantum Limit

For fixed geometry and vanishing interface decoherence,

$$\boxed{g_{\mu\nu} = g_{\mu\nu}^{fixed}, \quad \Gamma_I \to 0,}$$

standard quantum dynamics must be recovered:

$$\boxed{i\hbar \frac{\partial}{\partial t}\Psi = \widehat{H}\Psi.}$$

15.3 Standard Model Limit

If

$$\boxed{g_{\mu\nu} = g_{\mu\nu}^{fixed}, \quad \chi_I \to 0, \quad \zeta_a, \eta_H, y_{I,f} \to 0,}$$

then

$$\boxed{S_{TMI}^{min} \to S_{SM}.}$$

16 Levels of Claims and Limitations

To avoid mixing different types of claims, three levels are distinguished.

16.1 Level A: Internal Consequences

These are statements following from definitions. For example,

$$\boxed{g \Rightarrow \mathrm{Adm}_c(g)}$$

follows once $\mathrm{Adm}_c(g)$ is defined through $J_g^+(x)$. Likewise,

$$\boxed{\psi_I \in L^2(\mathrm{Adm}_c(g), \mathcal{B}_g, \mu_g) \Rightarrow d\mathbb{P}_{\Psi,g} = |\psi_I|^2 d\mu_g.}$$

16.2 Level B: Phenomenological Hypotheses

These are testable physical assumptions:

$$\boxed{\Gamma_I = \kappa_I |c_1|^2 |c_2|^2 d_I^2([g_1], [g_2]),}$$

$$\boxed{T_{\mu\nu}^{dark} \sim T_{\mu\nu}^{\chi},} \qquad \boxed{\Delta_a(E) = \zeta_a \frac{\chi_I^2(E)}{M_I^2}.}$$

They must not be presented as established facts of nature.

16.3 Level C: Open Problems

The following remain open: the full fundamental measure μ_G in the continuum limit, the rigorous transition $N \to \infty$, the fundamental operator $\widehat{\mathcal{H}}_{TMI}$, the complete solution of the problem of time, the derivation of $SU(3) \times SU(2) \times U(1)$ as a unique low-energy limit, the derivation of exactly three generations, numerical masses, CKM and PMNS matrices, and concrete experimental constraints.

17 Main Theorem of the Regularized Physical Layer

Theorem. Let a regularized space of physical geometries be given by

$$\boxed{\mathfrak{G}_N = \mathcal{G}_N / \operatorname{Diff}_N}$$

with measure μ_G^{cyl} and interface distance $d_{I,N}$. Let

$$\boxed{\mathcal{H}_G^{eff} = L^2(\mathfrak{G}_N, \mathcal{B}_N, \mu_G^{cyl})}$$

and let the interface decoherence coefficient be

$$\boxed{\kappa_I = \frac{m_I c^2}{\hbar} \mathcal{K}_I.}$$

If the open dynamics of geometry is

$$\boxed{\frac{d\rho_G}{dt} = -\frac{i}{\hbar}[\widehat{H}_G^{eff}, \rho_G] - \frac{1}{2}\kappa_I[\widehat{D}_I, [\widehat{D}_I, \rho_G]],}$$

then off-diagonal elements between distinct geometric classes are suppressed with rate

$$\boxed{\Gamma_I([g_1],[g_2]) = \kappa_I d_{I,N}^2([g_1],[g_2]).}$$

Consequently,

$$\boxed{\rho_G([g_1],[g_2]) \to 0 \quad ([g_1] \neq [g_2])}$$

and an effective classical geometry $g_{\mu\nu}^{cl}$ emerges.

Meaning of the theorem:

$$\boxed{\text{classical geometry is the interface-decohered limit of a quantum distribution of geome}}$$

18 Conclusion

The physical layer of the Theory of Manifold Interfaces is brought in this paper to the status of a regularized effective interface-field model. Its central formula is

$$\boxed{g \Rightarrow \operatorname{Adm}_c(g) \Rightarrow \mathcal{H}_I(g) \Rightarrow \Psi_I \Rightarrow \operatorname{Dist}_{\text{poss}}(\Psi_I; g) \Rightarrow \chi_I \Rightarrow \operatorname{Struct}_I.}$$

The extended architecture is

$$\boxed{(g_{\mu\nu}, \Psi, \chi_I, \mathbb{A}_I, \mathcal{G}_I) \Rightarrow (QG, SM, Gen, Dark, BH, ToE).}$$

In this form, TMI-Physics has a mathematical framework, a physical action, equations of motion, regimes, limiting cases, and an observable matrix. It is not a completed fundamental theory. Its correct status is

$$\boxed{TMI\text{-}Physics = \text{a regularized effective interface-field research program.}}$$

A Unified Register of Notation

Symbol	Meaning
$\mathcal{M}_U$	spacetime of the Universe
$g_{\mu\nu}$	Lorentzian metric
$T_x\mathcal{M}_U$	tangent space at x
$\mathcal{C}^+_{I,g}(x)$	future interface causal cone
$J^+_g(x)$	causal future of x
$\mathrm{Adm}_c(g)$	set of causally admissible transitions
$\mathcal{B}_g$	σ-algebra of measurable subsets
μ_g	effective or regularized measure
$\mathcal{H}_I(g)$	Hilbert space of interface possibilities
Ψ_I, ψ_I	interface quantum state and wave function
$\mathrm{Dist}_{\mathrm{poss}}$	distribution of possibilities
χ_I	interface field
Struct_I	structured physical regime
Γ_I	interface decoherence rate
κ_I	strength coefficient of the interface channel
d_I	interface distance between geometries
$\mathbb{A}_I$	universal interface connection
$\mathbb{F}_I$	universal interface curvature
$\mathcal{G}_I$	generation interface operator
$\Delta\Gamma_I$	excess interface decoherence
$w_I(z)$	equation-of-state parameter of interface dark energy
ρ_{corr}	correlation contribution to black-hole radiation
$\Delta_a(E)$	interface correction to gauge couplings

B Logical Order of Derivation

The definitions introduce $(\mathcal{M}_U, g)$, $\mathrm{Adm}_c(g)$, $\mathcal{H}_I(g)$, $\mathbb{P}_{\Psi,g}$, χ_I, and Struct_I. The postulates of the physical layer are

$$\boxed{P_1: \ g \Rightarrow \mathrm{Adm}_c(g),} \qquad \boxed{P_2: \ \Psi_I \Rightarrow \mathrm{Dist}_{\mathrm{poss}}(\Psi_I; g),} \qquad \boxed{P_3: \ \chi_I \Rightarrow \mathrm{Struct}_I\,.}$$

Internal consequences follow from these definitions. Phenomenological hypotheses specify testable relations: $\Delta\Gamma_I = \kappa_I x_I$, $T^{dark}_{\mu\nu} \sim T^{\chi}_{\mu\nu}$, $\rho_{rad} = \rho_{thermal} + \rho_{corr}$, and $\Delta_a(E) = \zeta_a \chi_I^2(E)/M_I^2$.

References

[1] Salkutsan Aleksey, *Theory of Manifold Interfaces: An Axiomatic Conceptual-Mathematical Framework of Interfaciality, Meaning, and Cognition*, Zenodo. https://doi.org/10.5281/zenodo.19940675.

[2] Salkutsan Aleksey, *Mathematical Abstraction of the Theory of Manifold Interfaces*, Zenodo. https://doi.org/10.5281/zenodo.19978895.

[3] Salkutsan Aleksey, *Theory of Manifold Interfaces. A Cautious Interface Approach to Quantum Gravity*, Zenodo. https://doi.org/10.5281/zenodo.19988505.

[4] Salkutsan Aleksey, *Theory of Manifold Interfaces TMI-QG-2.0: A Minimal Predictive Model of Interface-Induced Geometric Decoherence*, Zenodo. https://doi.org/10.5281/zenodo.20013976.

[5] Salkutsan Aleksey, *Black Holes in the Theory of Manifold Interfaces*, Zenodo. https://doi.org/10.5281/zenodo.20025226.

[6] Salkutsan Aleksey, *The Dark Sector as an Interface Field. A Module of the Theory of Manifold Interfaces*, Zenodo. https://doi.org/10.5281/zenodo.20025682.

[7] Salkutsan Aleksey, *Interface-Field Architecture for the Unification of Fundamental Interactions*, Zenodo. https://doi.org/10.5281/zenodo.20025731.

[8] A. Einstein, *Die Feldgleichungen der Gravitation*, Sitzungsberichte der Preussischen Akademie der Wissenschaften, 1915.

[9] R. M. Wald, *General Relativity*, University of Chicago Press, 1984.

[10] C. W. Misner, K. S. Thorne, J. A. Wheeler, *Gravitation*, W. H. Freeman, 1973.

[11] S. Weinberg, *The Quantum Theory of Fields*, Vol. 1, Cambridge University Press, 1995.

[12] M. E. Peskin and D. V. Schroeder, *An Introduction to Quantum Field Theory*, Addison-Wesley, 1995.

[13] J. D. Bekenstein, *Black holes and entropy*, Physical Review D, 7(8), 1973.

[14] S. W. Hawking, *Particle creation by black holes*, Communications in Mathematical Physics, 43, 1975.

[15] W. H. Zurek, *Decoherence, einselection, and the quantum origins of the classical*, Reviews of Modern Physics, 75, 2003.

[16] M. Schlosshauer, *Decoherence and the Quantum-to-Classical Transition*, Springer, 2007.

[17] Planck Collaboration, *Planck 2018 results. VI. Cosmological parameters*, Astronomy & Astrophysics, 641, A6, 2020.